MIL-C-83195A
25 January 1980
SUPERSEDING
MIL-C-83195 (USAF)
6 December 1968

MILITARY SPECIFICATION

COVERALLS, FLYER'S, ANTI-EXPOSURE CWU-21A/P

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of continuous wear anti-exposure coveralls that are designated CWU-21A/P.

1.2 Sizes. The coveralls shall be furnished in the following sizes as specified (see 6.2):

Size 1 (32-36) - Small short
Size 2 (32-36) - Small regular
Size 3 (32-36) - Small long
Size 4 (37-40) - Medium short
Size 5 (37-40) - Medium regular
Size 6 (37-40) - Medium long
Size 7 (41-44) - Large short
Size 8 (41-44) - Large regular
Size 9 (41-44) - Large long
Size 10 (45-47) - Extra large short
Size 11 (45-47) - Extra large regular
Size 12 (45-47) - Extra large long.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276	Thread, Cotton
L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
DDD-L-20	Labels: for Clothing, Equipage, and Tentage (General Use)
PPP-B-636	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-W-4088	Webbing, Textile, Woven Nylon
MIL-R-6130	Rubber, Cellular, Chemically Blown
MIL-C-7020	Cloth, Parachute, Nylon
MIL-C-19002	Cloth, Coated, and Tape, Coated Cloth-Polychloroprene on Nylon, Pneumatic Life Preserver
MIL-F-21840	Fastener Tapes, Hook and Pile, Synthetic

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-751	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-831	Test Reports, Preparation of

DRAWINGS

AIR FORCE

68J223	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 1
68J224	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 2
68J225	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 3
68J226	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 4
68J227	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 5
68J228	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 6
68J229	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 7
68J230	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 8
68J231	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 9
68J232	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 10
68J233	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 11
68J234	Patterns - Coveralls, Flying, Anti-Exposure CWU-21A/P, Size 12
79130	Patterns, Neck Seal - Coveralls, Flying, Anti-Exposure CWU-21A/P

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 4.4 and 6.2), the contractor shall furnish samples for first article inspection and approval.

3.2 Materials. Materials that are not covered by specifications or drawings or that are not specifically described herein shall be the best quality, as lightweight as practicable, and entirely suitable for the purpose intended. Except for the metallic parts and any materials furnished by the Government or

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unless otherwise specified herein, the materials used in the construction of the coveralls shall have been manufactured not more than 12 months prior to the date of delivery of the coveralls.

3.2.1 Basic fabric. The basic fabric shall be cotton ventile, 8-ounce, 36-inch wide, sage green USAF color shade No. 1509 and will be furnished by the Government (see 6.2).

3.2.2 Neoprene rubber. The material for the neck seal shall be neoprene rubber, with one side skin and the other side nylon; shall be 1/8 inch thick; shall be black; and shall conform to type II, grade A or B, condition soft of MIL-R-6130. The outside shall be skin, with the nylon on the inside.

3.2.3 Fastener tape. The hook and the pile fastener tape shall be 5/8 inch wide and shall conform to type II, class 1 of MIL-F-21840. The color of the hook and the pile fastener tape shall approximately match the color of the basic fabric.

3.2.4 Seam tape. The seam tape shall be a chloroprene-coated cloth and shall consist of layers of nylon cloth, polychloroprene coating, and calendered uncured gum polychloroprene coating that are specified in table I. The nylon cloth shall conform to type I of MIL-C-7020. The polychloroprene coating compound shall conform to MIL-C-19002. The calendered uncured polychloroprene gum coating shall be protected by a suitable liner that is capable of free separation without affecting the adhesion or cementability of the gum coating. The edges of the seam tape shall be smooth, not pinked. The age of the seam tape shall be as specified in MIL-C-19002. The seam tape shall be 3/4 inch wide. The color of the seam tape shall approximately match the color of the basic fabric.

TABLE I. Seam tape.

Layer No.	Materials	Weight (Ounces Per Square Yard) 1/
1	Polychloroprene coating	1.25
2	Bias-cut nylon cloth	1.10
3	Polychloroprene coating	1.25
4	Calendered uncured polychloroprene gum coating	2.50

1/ The seam tape shall weigh 6 ounces (plus or minus 0.75 ounce) per square yard.

3.2.5 Webbing. The protective flap for the entrance slide fastener shall be made of webbing. The webbing shall conform to type XII, untreated of MIL-W-4088 and shall be USAF color shade No. 1511 (see 6.3).

3.2.6 Cement. The cement shall be a chloroprene liquid cement and shall approximately match the color of the basic fabric. When the solvent in the cement evaporates, a flexible film shall remain. The film shall have suitable tackiness, strength, aging, and working qualities for the intended purpose.

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3.2.7 Thread. The thread for all sewing operations shall conform to type IC2; ticket No. B, 3 ply of V-T-276 and shall match the color of the basic fabric. The thread shall have at least good colorfastness to light and laundering.

3.2.8 Slide fasteners. The relief portal slide fastener shall conform to Style 1731, type I (closed both ends), size 84 (manufactured by Talon, Inc., Meadville, Pennsylvania), or equal. The entrance slide fastener shall conform to Style 1701, type II (open one end), size 84 (manufactured by Talon, Inc., Meadville, Pennsylvania), or equal. A thong that is made of black cotton tape and that is 3/8 inch wide and 2/32 inch (plus or minus 1/64 inch) thick shall be looped through and securely knotted to the stirrup ring at the center of the webbing length. The knot shall be tied as close as possible to the doubled thong. The length of the thong before looping shall not exceed 4 inches. The tape for the slide fasteners shall be black and shall be made of neoprene-coated nylon fabric. The dimensions of the slide fasteners shall conform to tables II and III. The seal blocks of the slide fasteners shall not rupture when subjected to a minimum load of 50 pounds. The slide fasteners shall not leak.

TABLE II. Dimensions of slide fastener tape.

Dimensions of Slide Fastener Tape	Requirements (Inches)
Overall width (with chain closed) 1/	2.00 (plus or minus 0.062)
Top and bottom extensions	1.00 (plus or minus 0.125)
Thickness (except for reinforced ends)	0.022 (plus or minus 0.003)
Thickness of reinforced ends	0.060 (plus or minus 0.010)

1/ The overall width shall include the chain.

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TABLE III. Length of chain.

Length of Chain in Inches (Plus or Minus 1/8)

Sizes of Coveralls	Entrance Slide Fastener	Relief Portal Slide Fastener
Size 1	22-1/2	6
Size 2	24	6
Size 3	25-1/2	6
Size 4	23-1/2	6
Size 5	25-1/4	6
Size 6	27	6
Size 7	24-3/4	6
Size 8	26-1/4	6
Size 9	27-3/4	6
Size 10	25-3/4	6
Size 11	27-1/4	6
Size 12	29	6

3.3 Design. The coveralls shall be a one-piece garment with a water impermeable seal at the neck and with pressure-and-water-sealing slide fastener closures.

3.4 Construction. The coveralls shall be constructed in accordance with table IV; however, the manufacturer will not be required to follow the exact sequence of operations as listed herein.

3.4.1 Stitches, seams, and stitchings. Stitches, seams, and stitchings used in the construction of the coveralls shall conform to FED-STD-751. Each row of stitching shall be straight and parallel to the seam edges. A tolerance of plus or minus 1/32 inch will be allowed in the nominal distance of stitches from the edge on the joining seams. The thread tension shall be maintained so that stitching will not be loose and the thread lock will be imbedded in the center of the materials being sewn together. Skipped stitches and run offs shall be repaired using stitch type 301. Thread breaks shall be secured by stitching back of the break not less than 1/2 inch. The ends of continuous lines of stitchings (except for stitching on labels) that are not caught in other seams or stitchings shall overlap not less than 1/2 inch. The ends of stitching on labels shall overlap not less than three stitches.

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TABLE IV. Manufacturing operations.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
1	Join fronts			
	(a) The right front and the left front shall be joined, starting at the crotch and ending at the notch for the entrance slide fastener, with the right front overlapping the left front, with a double-lapped seam, 3/16-inch gage.	401	LSc-2	8-10
2	Join backs			
	(a) The right back and the left back shall be joined, with the right back overlapping the left back, with a double-lapped seam, 3/16-inch gage.	401	LSc-2	8-10
3	Join shoulders			
	(a) The assembled fronts and the assembled backs shall be joined at the shoulders, with the fronts overlapping the backs, with a double-lapped seam, 3/16-inch gage.	401	LSc-2	8-10
4	Join undersleeve to topsleeve			
	(a) The topsleeve and the undersleeve shall be joined at the back, with the topsleeve overlapping the undersleeve, with a double-lapped seam, 3/16-inch gage.	401	LSc-2	8-10
5	Install seam tape			
	(a) The seam tape shall be cemented as specified in 3.4.2 on the inside of the coveralls so that the seam tape will cover the front seam, the back seam, the shoulder seams, and the back sleeve seams.			

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TABLE IV. Manufacturing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
6	Join sleeves to armholes			
	(a) The sleeves shall be joined to the armholes, with the body of the coveralls overlapping the sleeves, with a double-lapped seam, 3/16-inch gage.	401	LSc-2	8-10
	(b) The seam tape shall be cemented as specified in 3.4.2 on the inside of the coveralls so that the seam tape will cover the armhole seams.			
7	Install relief portal slide fastener			
	(a) The relief portal slide fastener shall be positioned on the inside of the coveralls so that the chain will be toward the outside of the coveralls, the center line of the chain will be along the slide fastener line shown on the pattern, and the slide fastener will close toward the left (as worn). The relief portal slide fastener shall be stitched around the sides and the ends (see figure 1) to the inside of the coveralls with one row of continuous stitching 1/16 to 3/32 inch from the edges on the sides and 1/2 to 5/8 inch from the stop blocks on the ends.	301	LSb-2	8-10
	(b) The coveralls shall be turned and cut along the slide fastener line in the manner shown on figure 1. The fabric shall be folded under at the designated lines and stitched along the turned edge in accordance with figure 2.	301	LSb-2	8-10
	(c) The seam tape shall be cemented as specified in 3.4.2 and figure 2 on the inside of the coveralls over the relief portal slide fastener so that the edge of the seam tape that is closer to the center line of the slide fastener will extend over the inside row of stitching not less than 3/16 inch nor more than 1/4 inch.			

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TABLE IV. Manufacturing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
8.	Install entrance slide fastener			
	(a) The entrance slide fastener shall be positioned at the front and the neck area on the inside of the coveralls. The coveralls shall be notched in accordance with figure 3. The basic fabric shall be folded under at the designed lines in accordance with figure 3 and stitched along the turned edges.	301	LSb-2	8-10
	(b) The coveralls shall be turned. The entrance slide fastener shall be stitched around the sides and the bottom of the slide fastener 1/16 to 3/32 inch from the edges on the sides and 1/2 to 5/8 inch from the stop block at the bottom.	301	LSb-2	8-10
	(c) The seam tape shall be cemented as specified in 3.4.2 and figure 2 to the inside of the coveralls to cover the entrance slide fastener with the edge of the seam tape that is closer to the center line of the entrance slide fastener extending over the inside row of stitching not less than 3/16 inch nor more than 1/4 inch.			
9	Make protective flap			
	(a) A strip of the seam tape with the backing left on shall be joined lengthwise along one edge to the protective flap, with the seam tape overlapping the protective flap 1/2 inch, with a single row of stitching (see figure 4). The ends of the stitching shall be backstitched not less than 1/2 inch.	301	LSa-1	8-10

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No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
	Note: The strip of seam tape shall be 1/2 inch longer on each end than the protective flap. The protective flap shall be 2-1/2 inches longer than the entrance slide fastener. The ends of the protective flap (webbing) shall be seared.			
	(b) Three pieces of the hook fastener tape that are 1 inch (plus or minus 1/8 inch) long shall be positioned on the outside of the protective flap so that one piece will be in the center of the protective flap and one piece will be 2 inches (plus or minus 1/2 inch) from each end of the protective flap and so that the 1-inch dimension of the hook fastener tape will be parallal and even with the edge of the protective flap (see figure 4). The three pieces of the hook fastener tape shall be joined on all four sides to the protective flap with a single row of stitching 1/16 inch (plus or minus 1/32 inch) from the edge.	301	SSaa-1	8-10
10	Install protective flap			
	(a) The backing shall be removed from the seam tape that is stitched to the protective flap. The seam tape that is stitched to the protective flap shall be cemented as specified in 3.4.2 over the entrance slide fastener seam tape (see figure 4) so that the edges will be even and the protective flap will be on the right side (as worn) of the entrance slide fastener.			
	Note: The protective flap in the finished coveralls shall extend from the joining seam of the neck seal to 1 inch (plus or minus 1/8 inch) below the stop block at the bottom of the entrance slide fastener.			

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TABLE IV. Manufacturing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
11	Attach hook fastener tape (a) The protective flap shall be smoothed out over the entrance slide fastener. The position for three pieces of the pile fastener tape that are 1 inch (plus or minus 1/8 inch) long shall be marked on the entrance slide fastener seam tape to assure that the protective flap will be smooth and cover the entrance slide fastener without distortion or interference with the operation of the slide fastener. The three pieces of the pile fastener tape shall be cemented as specified in 3.4.2 to the entrance slide fastener seam tape so that the 1-inch dimension of the pile fastener tape will be parallel to and even with the edge of the seam tape.			
12	Attach label (a) The label shall be positioned equidistant between the top and the center pieces of the hook fastener tape with the length dimension of the label parallel to the length dimension of the protective flap and with the printed side of the label to the outside of the coveralls and readable when facing the opened entrance slide fastener. The label shall be stitched around all four sides with a single row continuous stitching 1/8 inch from the edge.	301	LSbj-1	8-10
13	Join sides and undersleeves (a) The sides and the undersleeves shall be joined, with the fronts overlapping the backs, with a double-lapped seam, 3/16-inch gage.	401	LSc-2	8-10

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TABLE IV. Manufacturing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
	(b) The seam tape shall be cemented as specified in 3.4.2 to the inside of the coveralls so that the seam tape will cover the side and underarm sleeve seams.			
14	Join inseams			
	(a) The inseams shall be joined, with the front overlapping the back, with a double-lapped seam, 3/16-inch gage.	401	LSc-2	8-10
	(b) The seam tape shall be cemented as specified in 3.4.2 to the inside of the coveralls so that the seam tape will cover the inseams.			
15	Overedge stitch bottom of sleeves and legs			
	(a) The bottom raw edge of each sleeve and each leg shall be overedge stitched.	502 or 503	EFd-1	6-10
16	Attach neck seal			
	(a) The neck seal shall be cemented as specified in 3.4.2 to the neck of the coveralls with the coveralls overlapping the neck seal 1/2 inch (plus or minus 1/16 inch), with the extension of the neck seal on the right side (as worn), and with the notches of the neck seal matching the center back seam and the shoulder seam.			
	(b) The seam tape shall be cemented as specified in 3.4.2 to the outside of the coveralls so that the seam tape will extend onto the neck seal and the coveralls not less than 5/16 inch and will cover the raw edge of the neck cutout on the outside of the coveralls.			

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TABLE IV. Manufacturing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
	(c) The seam tape shall be cemented as specified in 3.4.2 to the inside of the coveralls so that the seam tape will extend onto the neck seal and the coveralls not less than 5/16 inch and will cover the raw edge of the neck seal on the inside of the coveralls.			
17	Dust taped seams (a) All the taped seams of the coveralls shall be dusted with zinc stearate or any other suitable powder to dull the exposed cemented areas along each seam tape. All residue of loose powder shall be removed from the seam tape and the adjoining fabric area by brushing with a medium hard bristle (not wire) brush.			

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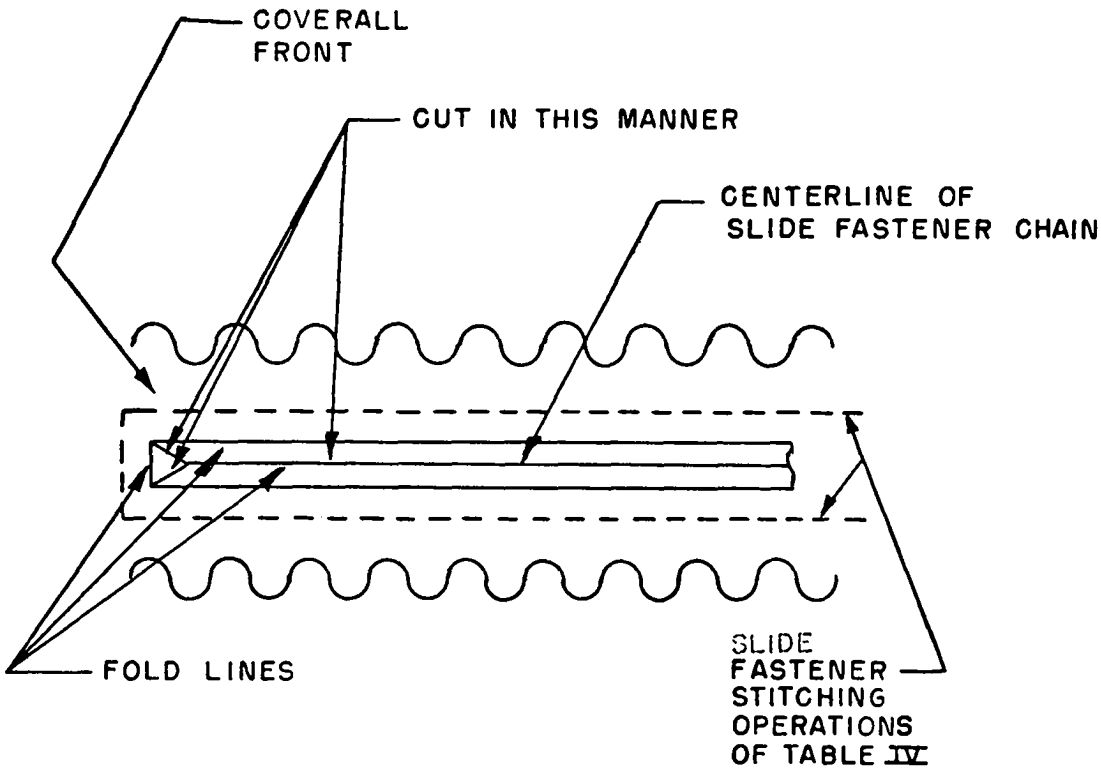
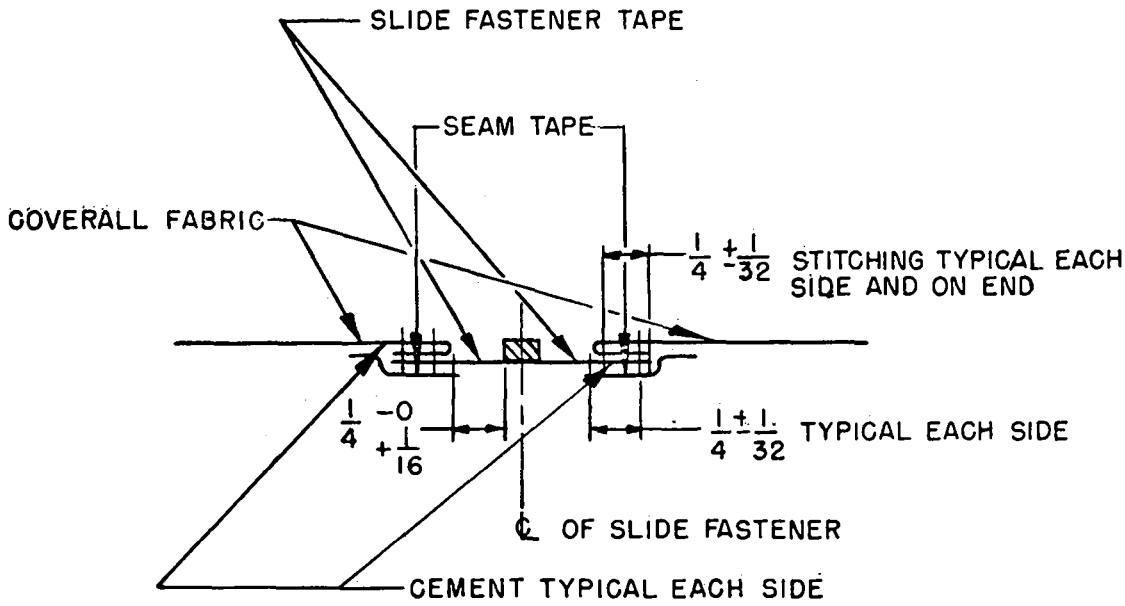


FIGURE 1. Installation of relief portal slide fastener.

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DIMENSIONS IN INCHES

FIGURE 2. Installation of slide fasteners.

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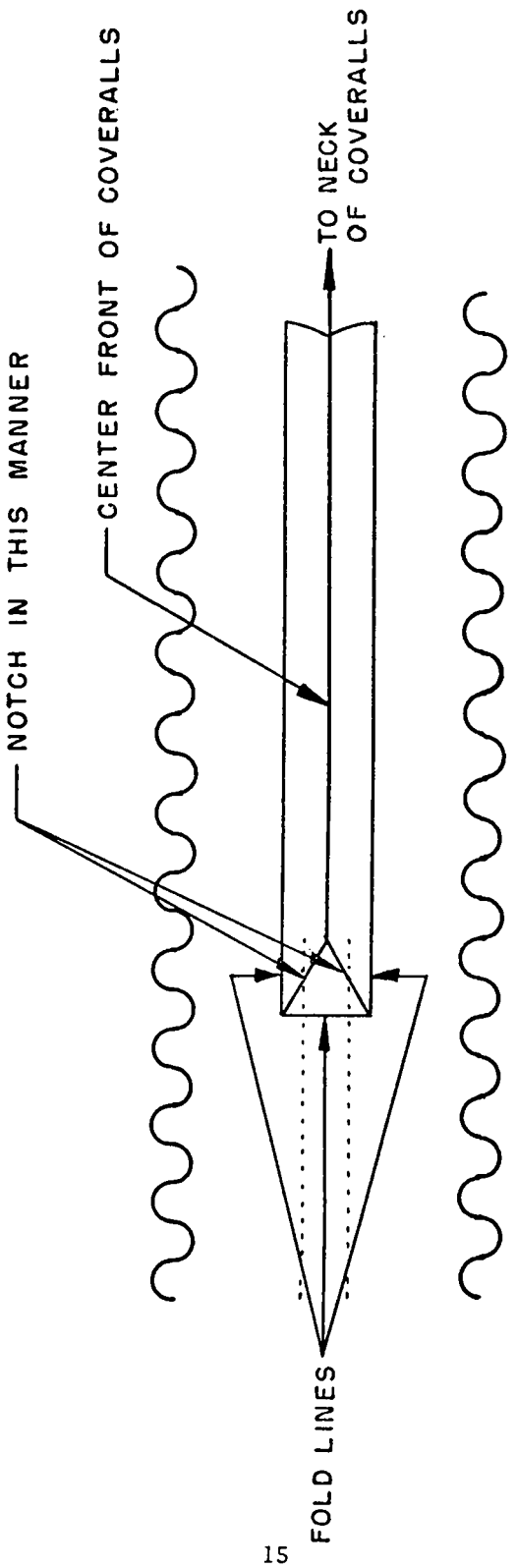
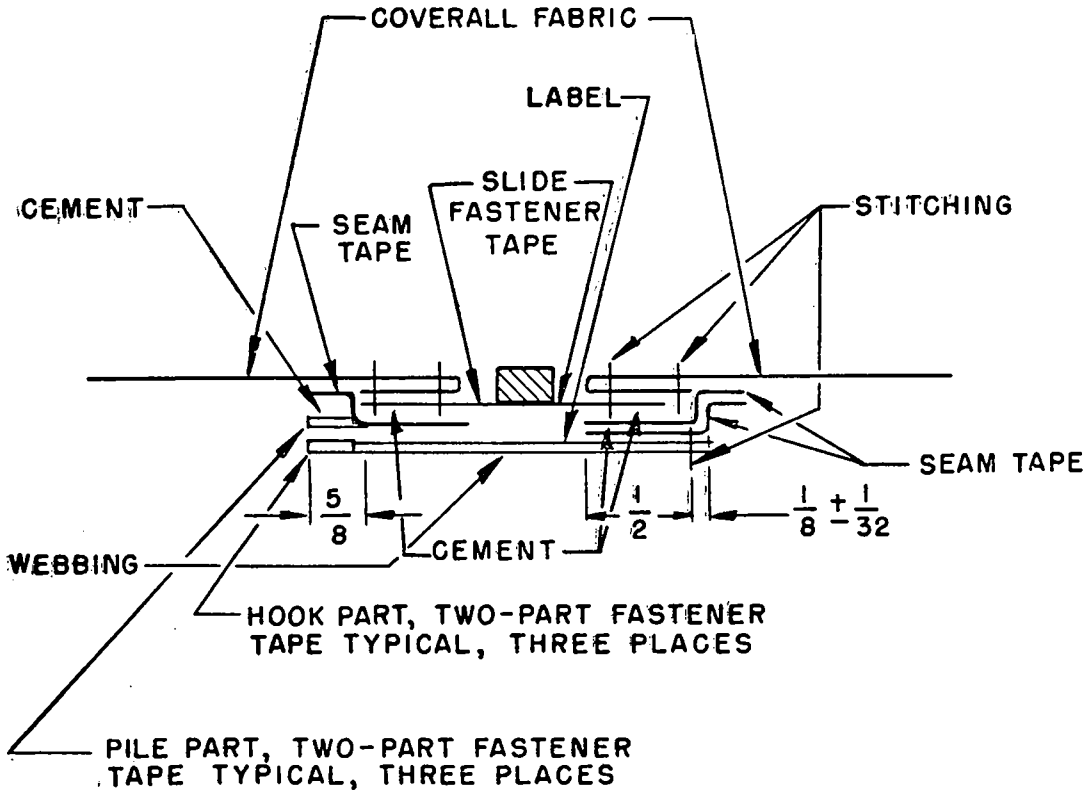


FIGURE 3. Notch for entrance slide fastener.

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FIGURE 4. Protective flap.

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3.4.2 Cementing operations. The seam tape shall be cemented on the inside of the coveralls to cover all stitching that perforates the coveralls and to cover the raw edge of the neck seal where it is attached to the coveralls. The seam tape shall be cemented on the outside of the coveralls to cover the raw edge of the neck cutout of the coveralls where it is attached to the neck seal. The stitched seams of the coveralls shall not be cleaned with solvent. The surface of the neck seal to which the cement is to be applied shall be thoroughly cleaned with a suitable solvent so that foreign matter including any dusting materials such as zinc stearate or talc will be removed to prevent contamination of the cement. Brushing with a stiff bristled brush will be permitted. Care shall be exercised to prevent prolonged exposure of the basic fabric, the neoprene rubber, and the thread to the solvent. The solvent shall have evaporated completely prior to the application of the cement and shall not leave any residue. All trapped air, channels, and wrinkles shall be removed from the cemented areas of the neck seal and of the stitched seams. All areas that are to be cemented shall be secured with the cement specified in 3.2.6 and in accordance with the recommendations of the manufacturer of the cement to achieve the optium bonding properties and an adhesive of not less than 5 pounds. Not less than two coats of cement shall be used on all areas that are to be cemented. All seam tape shall be applied without tension and with not less than 5/32 inch of the seam tape projecting beyond the folded seam edge or the inside row of double stitching used on the seam. The cement shall be not less than 1/8 inch nor more than 1/4 inch beyond the edges of the seam tape. The seam tape on the coveralls shall overlap a minimum of 3/4 inch at all joining points. Every effort shall be made to avoid tape end joints where the seam tape passes over the taped seams of the coveralls. Old cement or cement that has partly or completely polymerized shall not be used. Before containers are filled with fresh cement, the containers shall be freed of congealed cement. A fresh batch of cement shall be used at least every 8 hours during the manufacture of the coveralls.

3.4.3 Patching. Patching of fabrication defects shall not be permitted.

3.4.4 Cutting. The parts of the coveralls shall be cut, from the materials specified herein, in strict accordance with the pattern parts specified herein. No punch marks shall be made through the material in marking off the pattern parts.

3.5 Patterns. The coveralls shall conform to the applicable pattern parts that are referenced on Drawings 68J223 through 68J234 and 79130 and that are listed in table V. The patterns that are referenced on these drawings show directional lines, size and shape, notches, and markings for the proper assembly of the coveralls and shall be used as a guide for processing the manufacturer's working patterns. Unless otherwise specified herein, the seam allowance provided by the pattern parts is 3/8 inch. The seam allowance provided by the pattern part for the neck seal is 1/2 inch. The patterns referenced on these drawings and the manufacturer's working patterns shall be identical and shall not be altered in any way without prior approval of the procuring activity.

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TABLE V. Pattern parts.

Materials	Patterns Parts	Quantity
Basic fabric (see 3.2.1)	Right front	1
	Left front	1
	Back	2
	Topsleeve	2
	Undersleeve	2
Neoprene rubber (see 3.2.2)	Neck seal	1
Webbing (see 3.2.5)	Protective flap 1/	1

1/ A pattern part for the protective flap will not be required.

3.6 Identification of product. The label shall conform to type I, classes 1 and 2 of DDD-L-20 and shall have good colorfastness to laundering. The label shall be approximately 1-1/2 inches by 2-3/4 inches. The label shall contain the following information:

Coveralls, Flyer's, Anti-Exposure CWU-21A/P
Specification MIL-C-83195A
Manufacturer's Identification 1/
Contract No. 1/
Date of Manufacture 1/
Stock No. 1/
Size 1/

1/ The manufacturer shall include the applicable information.

3.7 Workmanship. The coveralls shall be uniform in quality and shall be free from irregularities or any defects that could adversely affect performance, reliability, or durability. Because the coveralls are for emergency and life support use, the importance of providing a product of uniformly excellent quality cannot be overemphasized.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsibility for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Certificate of compliance. A certificate of compliance that has been prepared by the manufacturer of the slide fasteners will be acceptable for the requirements specified in 3.2.8. If a certificate of compliance is furnished for any requirement specified herein, the Government shall have the right to check test items to determine the validity of the certification.

4.3 Classification of inspection. The inspection required by this specification is classifed as follows:

(a) First article inspection

(b) Quality conformance inspection.

4.4 First article inspection. The first article (see 3.1 and 4.4.1) shall be subjected to the first article inspection. The first article inspection shall consist of all of the inspection methods described under 4.6.

4.4.1 First article. The first article shall consist of three coveralls, each of a different size, and shall be representative of the materials, the components, the design, the construction, and the workmanship to be used in the production items. If a contractor is in continuous production of the coveralls covered by this specification from contract to contract or within a 12-month period, the submission of the first article on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article or the waiver of the first article by the procuring activity shall not preclude the requirements for the performance of the quality conformance inspection.

4.4.2 Inspection report. When required (see 4.2 and 6.2), three copies of an inspection report in the format specified in MIL-STD-831 shall be submitted with the first article.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the following:

(a) Individual inspection

(b) Sampling inspection.

4.5.1 Individual inspection. Each of the coveralls shall be subjected to the visual examination for critical and major defects specified in 4.6.1.1.

4.5.2 Sampling inspection. Samples that have been selected in accordance with inspection level II of MIL-STD-105 shall be subjected to the visual examination for minor defects specified in 4.6.1.2. The sample unit and the unit for the expression of the lot size for this sampling inspection shall be one of the coveralls. The acceptable quality level for this sampling inspection shall be 6.5 defects per 100 units.

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4.6 Inspection methods

4.6.1 Visual examination

4.6.1.1 Critical and major defects. The coveralls shall be visually examined for the critical and the major defects listed in table VI. Any critical or major defect shall be cause for rejection of the coveralls.

4.6.1.2 Minor defects. The coveralls shall be visually examined for the minor defects listed in table VI.

4.6.2 Seam tape adhesion. The sample shall be 3 inches wide and 8 inches long. The sample shall be an 8-inch-long piece of seam tape cemented over a stitched seam in the same manner as the seams of the coveralls except that 2 inches of the seam tape shall be left uncemented. The sample shall be allowed to cure before it is tested. After the sample has been allowed to cure, the adhesion of the seam tape of the sample shall be determined in accordance with method 5960 of FED-STD-191 and shall be not less than 5 pounds.

4.6.3 Slide fastener leakage. Before the slide fasteners are installed in the coveralls, the entrance slide fastener and the relief portal slide fastener shall be inspected for leakage as described herein. Each slide fastener shall be inserted into a suitable jig. An oil and liquid water free air pressure of 5 pounds per square inch shall be applied to the individual sections of the slide fastener. While this pressure is applied to the sealing side of the slide fastener for not less than 10 seconds, the reverse side of the slide fastener shall be immersed in water. Any evidence of leakage through the slide fastener from the side where the air pressure is applied to the side immersed in water shall be cause for rejection of the slide fastener. A white letter "W" shall be stamped on the tape of each slide fastener that has passed the slide fastener leakage test.

4.6.4 Seal block tearing strength. The apparatus described in method 5100 of FED-STD-191 shall be used to determine the tearing strength of the seal block of the opened entrance slide fastener. The fastener shall be opened not less than 4 inches from the seal block. The tape on each side of the fastener shall be inserted in the clamping blocks. The ends of the seal block shall be even with the end of the slide fastener opening and the start of the seal block. The opposite ends of the clamping blocks shall extend toward the center of the slide fastener. The tearing strength of the seal block shall be determine five times. The tearing strength of the seal block shall be the average of the five determinations. The seal block shall withstand a minimum load of 50 pounds without rupturing.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. A sheet of kraft paper shall be laid over the entire surface of each of the coveralls. The coveralls, including the kraft paper, shall be neatly folded in accordance with steps 1 through 5 of figure 5. The folded coveralls shall be inserted in a snug-fitting bag. The bag shall be made of polyethyelene that conforms to L-P-378 and shall be heat sealed. Prior to sealing the bag, all excess air shall be removed from the bag.

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TABLE VI. Classification of defects.

Defects	Critical	Major	Minor
Material defects and workmanship damages			
Hole, scissor or knife cut, tear, burn, abraded area, oil or soil mark, or any other weakening defects resulting from manufacturing process	X		
Any spot or stain such as solvents, cements, or powders that cannot be removed without damage to coverall fabric or components	X		
Hook and pile fastener tape			
Any hook or pile fastener tape not positioned as specified herein		X	
Any pile fastener tape not cemented properly or with less adhesion than specified		X	
Slide fasteners			
Any part of slide fastener bent, broken, or otherwise defective 1/	X		
Relief portal slide fastener not closing in direction specified		X	
Length of slide fastener not as specified	X		
Components and assembly			
Any defective component or defect that will seriously affect life saving characteristics of coveralls (unless otherwise classified herein)	X		
Any component not as specified or required operation improperly performed (unless otherwise classified herein)	X		
An part not cut in accordanc with pattern		X	
Any component not specified color			X
Cementing operations			
Cement on cloth surfaces along seam tape or attachment edges less than 1/8 inch or more than 1/4 inch			X
Location of any seam tape not as specified		X	

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TABLE VI. Classification of defects. - Continued

Defects	Critical	Major	Minor
Any seam tape applied in crooked manner for distance exceeding 1 inch but not exceeding 2 inches in length (less than 1 inch in length not to be classified as defect)			X
Any seam tape applied in crooked manner for distance exceeding 2 inches in length		X	
Loosening of seam tape from protective flap which is not repairable	X		
Loosening of seam tape from protective flap which is repairable		X	
Any supported seam tapes not lapped specified distance		X	
An damaged seam tape edges greater than 1/16 inch from outside edge			X
Neck seal attachment not covered on inner and outer sides or not covered with specified seam tape	X		
Slide fastener attachment not covered on inner side, not covered properly, or not covered with specified seam tape	X		
Seams and stitchings			
Thread ends not trimmed or loose thread ends not removed			X
Any seam or attachment of any component twisted, puckered, pleated, or caught in any unrelated operation or stitching		X	
Ends of stitching (except on label) that are not caught in other seams or stitchings not overlapped at least 1/2 inch		X	
Ends of stitching on label not overlapped at least three stitches			X
Thread breaks not secured by stitching back of break not less than 1/2 inch			X
Skipped stitches or run off not repaired as specified herein			X
Any stitching irregular or unevenly gaged on major portion of seam			X

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TABLE VI. Classification of defects. - Continued

Defects	Critical	Major	Minor
Seam or stitch type not as specified		X	
Loose tension resulting in loose seam		X	
Tight tension resulting in stitches breaking when normal pull is applied 2/		X	
Stitches per inch on major portion of seam one or two stitches per inch more or less than specified stitches per inch			X
Stitches per inch on major portion of seam more than two stitches per inch more or less than specified stitches per inch		X	
Label			
Label missing, incorrect, incomplete, illegible, or insecurely attached			X

1/ The slide fastener shall be checked for proper functioning by opening and closing each slide fastener a minimum of three times along its full length.
2/ Puckering is evidence of tight tension. If puckering is evident, normal pull shall be exerted in the lengthwise direction of the seam or stitching.

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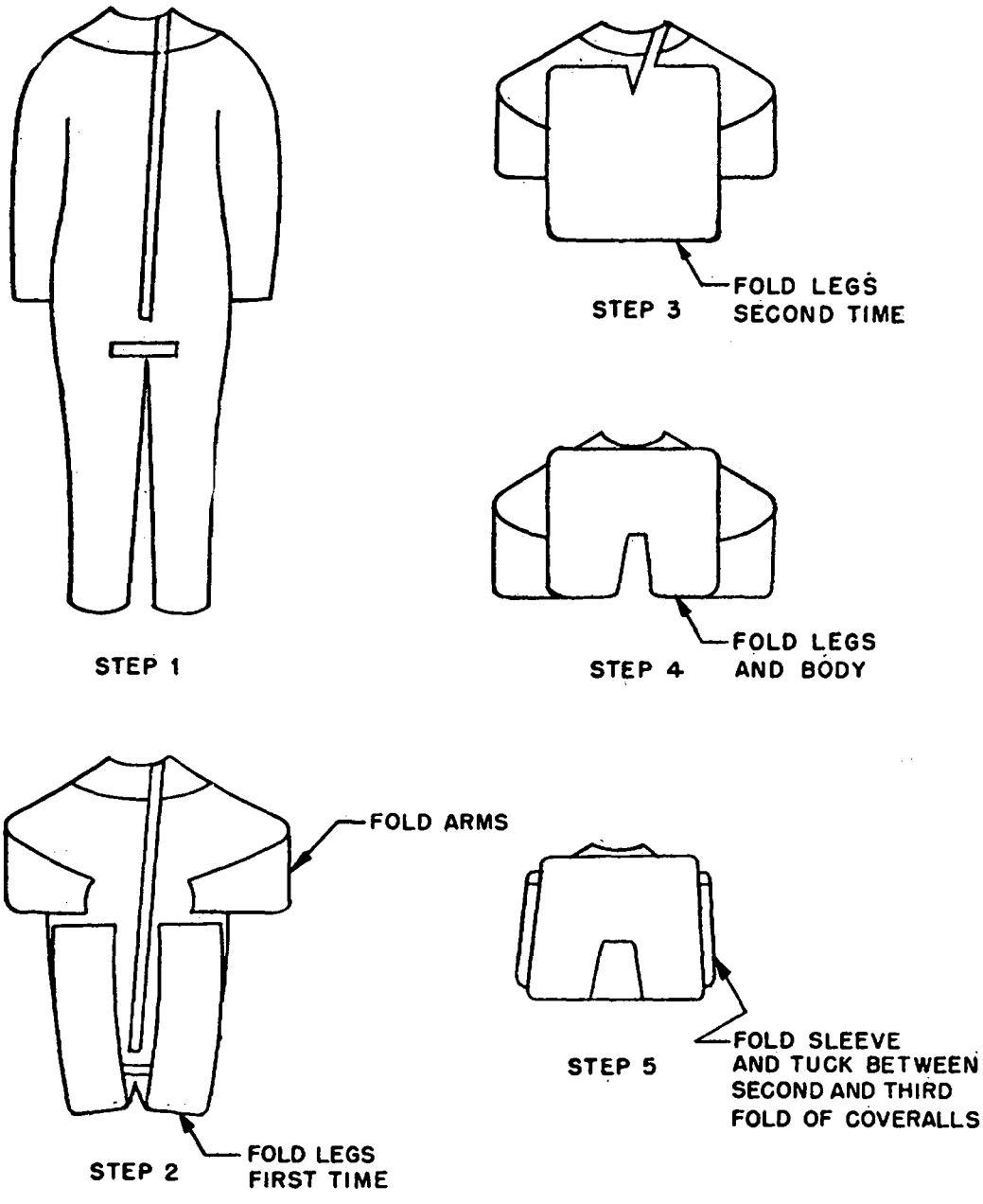


FIGURE 5. Folding of coveralls.

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5.1.1.1 Intermediate packaging. Four coveralls of one size only, preserved and packaged as specified in 5.1.1, shall be placed in a snug-fitting box that conforms to type optional, class domestic, grade 275 or better of PPP-B-636. The closure of the box shall be in accordance with the appendix of the box specification.

5.1.2 Level C. The coveralls shall be preserved and packaged to afford adequate protection, against deterioration and physical damage, during shipment from supply source to the first receiving activity. This level of preservation and packaging may conform to the supplier's commercial practice provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Sixteen coveralls of one size only, preserved and packaged as specified, shall be packed in a box that conforms to class weather-resistant, style RSC or PPP-B-636. The box shall be closed and reinforced in accordance with the appendix of the box specification. If the quantity of coveralls of the same size is less than the quantity of coveralls required per box, mixed sizes (see 5.3.1) may be packed within the same box.

5.2.2 Level B. Sixteen coveralls of one size only, preserved and packaged as specified, shall be packed in a box that conforms to class domestic, style RSC of PPP-B-636. The box shall be closed and reinforced in accordance with the appendix of the box specification. If the quantity of coveralls of the same size is less than the quantity of coveralls required per box, mixed sizes (see 5.3.1) may be packed within the same box.

5.2.3 Level C. The coveralls shall be packed to afford adequate protection, at the lowest rate, against damage during direct shipment from the supply source to the first receiving activity. The boxes shall conform to the applicable rules of carriers and regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract, each unit package, intermediate package, and shipping container shall be marked in accordance with MIL-STD-129.

5.3.1 Mixes sizes. Each shipping container that is packed with mixed sizes (see 5.2.1 and 5.2.2) shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "mixed sizes" plainly stamped or printed thereon. Under these words shall be legibly stamped or printed the quantities and sizes of the coveralls therein.

6. NOTES

6.1 Intended use. The coveralls covered by this specification are intended to have wristlets and socks of the correct size custom fitted to the individual in the field for continuous wear under outer garments by aircrewmembers on overwater flights to protect the wearer from exposure to cold water, wind, spray, and rain while marooned in a life raft. The coveralls are intended to be used in conjunction with the CWU-22/P underwear and the CWU-23/P liner.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Size required (see 1.2)

(c) First article requirements (see 3.1, 4.4, 4.4.1, and 4.4.2)

(d) Government furnished material (see 3.2.1)

(e) Selection of applicable levels of preservation and packaging and packaging (see 5.1 and 5.2).

6.3 Color samples. Color samples can be obtained from the procuring activity or as directed by the contracting officer.

6.4 Figures. Figures 6 and 7 show general design of the coveralls and are furnished for information only.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes, with respect to the previous issue, due to the extensiveness of the changes.

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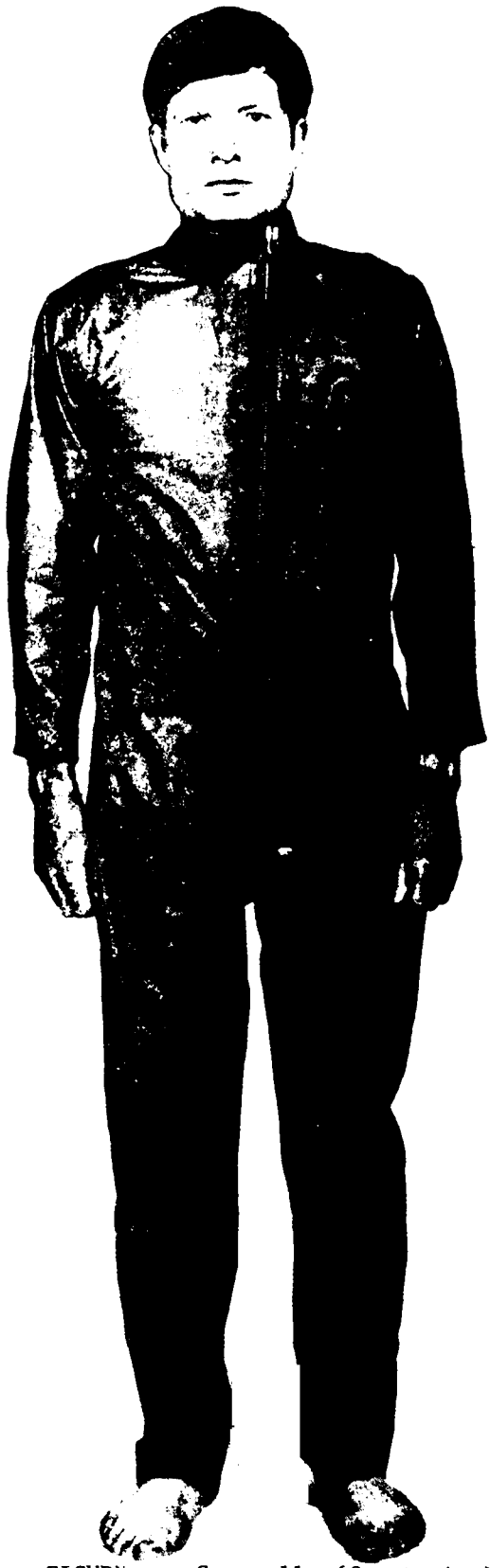


FIGURE 6. Coveralls (front view).

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FIGURE 7. Coveralls (rear view).

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